Phillip Lippe

PhD student fascinated by research in artificial intelligence and deep learning, especially causality and representation learning. Looking for Research Scientist positions based in Europe and US with starting dates from October 2024 on.

Gender Male   Age 25 years   Nationality German   Languages German (native), English (C1)   Amsterdam, Netherlands
@ phillip.lippe@googlemail.com   Phone number on request   1095MB Amsterdam
phlippe.github.io   linkedin.com/in/phillip-lippe   github.com/phlippe   @phillip_lippe

EXPERIENCE

Student Researcher
Google DeepMind
Aug 2023 – Nov 2023   Amsterdam, Netherlands

- Research in generative multimodal pretraining with positive transfer across modalities; hosted by Mostafa Dehghani
- Parallelized training of models with up to 4 billion parameters
- Results internally used in the Gemini foundation model project

Research Intern
Microsoft Research
Mar 2023 – May 2023   Amsterdam, Netherlands

- Research in neural PDE solvers for scientific simulations like fluid dynamics; hosted by Johannes Brandstetter
- Results published in *PDE-Refiner: Achieving Accurate Long Rollouts with Neural PDE Solvers* at NeurIPS 2023 (pdf)

Student Research Intern
Daimler AG, Mercedes-Benz
Dec 2016 – Mar 2017
Apr 2018 – Sep 2018   Stuttgart, Germany

- Research in deep learning for real-time multi-class object detection in complex urban traffic scenes (autonomous driving)
- Bachelor Thesis: *Hierarchical Multi-label Object Detection of Rare Classes for Autonomous Driving* (pdf)

Student Research Intern
Mercedes-Benz Research and Development NA
May 2017 – Aug 2017   Sunnyvale, United States

- Research in deep generative models for predicting agent behavior in traffic scenarios for autonomous driving (report)
- Performed agile software development with Scrum

PROJECTS

Google Developer Expert, Machine Learning
- Member of the ML GDE program for expertise and public content sharing in JAX+Flax since August 2022

Tutorials
- Created and taught a series of implementation tutorials in PyTorch and JAX for various Deep Learning topics (website, ~40k page visits per month, >1.8k GitHub stars). Part of the DL courses at UvA and official tutorials of PyTorch Lightning

Teaching assistant
- Teaching assistant (TA) for the graduate courses Deep Learning, NLP 1, FAIR, IR 1, and Advanced Topics in Computational Semantics at the University of Amsterdam (2019-2023)
- Head TA for ASCI PhD Course on Computer Vision 2022 (link)

Summer schools
- Participated in the Oxford ML Summer School 2021, and the CIFAR DL and Reinforcement Learning Summer School 2021

EDUCATION

PhD, Artificial Intelligence
University of Amsterdam, QUVA lab
Sep 2020 – exp. Sep 2024   Amsterdam, Netherlands

- Supervisors: dr. Efstratios Gavves and dr. Taco Cohen
- My research focuses on the intersection of causality and deep learning, in particular causal representation learning (abstract)
- ELLIS PhD; collaboration with Qualcomm AI Research

Master of Science, Artificial Intelligence
University of Amsterdam
Sep 2018 – Aug 2020   Amsterdam, Netherlands

- Courses on AI, including ML, DL, NLP, IR, CV, and RL
- Thesis on "Categorical Normalizing Flows" (publication), applied for permutation-invariant graph/molecule generation
- Final GPA: 9.5, cum laude (Dutch grading system)

Bachelor of Engineering, Computer Science
Baden-Wuerttemberg Cooperative State University
Oct 2015 – Sep 2018   Stuttgart, Germany

- Cooperative study program with the specialization in IT Automotive and autonomous driving, in cooperation with Daimler AG/Mercedes-Benz R&D
- Final GPA: 1.0 (German grading system)

Pre-Studies, Mathematics/Computer Science
Ruhr University Bochum
Apr 2012 – Sep 2014   Bochum, Germany

- Completing university courses as high school student, including Introduction to Software Engineering I & II, Linear Optimization, and Programming for Mathematicians

AWARDS

- Won 4th place in the NeurIPS 2020 challenge "Hateful Memes" of Facebook AI (paper)
- Best CS undergraduate student at the Baden-Wuerttemberg Cooperative State University 2018
- Awarded SchülerUni Scholarship 2012/13 for taking university courses on Computer Science/Mathematics during high school

SKILLS

DL Frameworks: PyTorch (2018-present, R&T), JAX/Flax (2021-present, R&T), Tensorflow (2017-2018, R), Caffe (2016-2017, R)   (R - Research, T - Teaching)

Programming languages: Python (2016-present), HTML/PHP/SQL (2012-present), Java (2012-2019), MATLAB (2016-2019), C (2015-2017)

Additional skills: SLURM cluster computing (2018-present), git (2015-present), Docker (2017-present)

SELECTED PUBLICATIONS

Conferences

- Davide Talon, Phillip Lippe, Stuart James, Alessio Del Bue, Sara Magliacane: **Towards the Reusability and Compositionality of Causal Representations**. Third Conference on Causal Learning and Reasoning (CLeaR), 2024 (link) **[Oral]**
- Phillip Lippe, Bastiaan S. Veeling, Paris Perdikaris, Richard E. Turner, Johannes Brandstetter: **PDE-Refiner: Achieving Accurate Long Rollouts with Neural PDE Solvers**. Thirty-seventh Conference on Neural Information Processing Systems (NeurIPS), 2023 (link). **[Spotlight]**
- Sindy Löwe, Phillip Lippe, Francesco Locatello, Max Welling: **Rotating Features for Object Discovery**. Thirty-seventh Conference on Neural Information Processing Systems (NeurIPS), 2023 (link). **[Oral]**
  — *Own contributions:* Advised in model development, experiment design and model optimization. Contributed to writing.
- Phillip Lippe, Sara Magliacane, Sindy Löwe, Yuki M. Asano, Taco Cohen, Efstratios Gavves: **BISCUIT: Causal Representation Learning from Binary Interactions**. The 39th Conference on Uncertainty in Artificial Intelligence (UAI), 2023 (link). **[Spotlight]**
- Phillip Lippe, Sara Magliacane, Sindy Löwe, Yuki M. Asano, Taco Cohen, Efstratios Gavves: **Causal Representation Learning for Instantaneous and Temporal Effects in Interactive Systems**. International Conference on Learning Representations (ICLR), 2023 (link).
- Adeel Pervez, Phillip Lippe, Efstratios Gavves: **Differentiable Mathematical Programming for Object-Centric Representation Learning**. International Conference on Learning Representations (ICLR), 2023 (link).
  — *Own contributions:* Proposed application to object-centric learning. Advised in model development and experimental setups. Contributed to writing.
- Adeel Pervez, Phillip Lippe, Efstratios Gavves: **Scalable Subset Sampling with Neural Conditional Poisson Networks**. International Conference on Learning Representations (ICLR), 2023 (link).
  — *Own contributions:* Advised in model development and experimental setups. Contributed to writing.
- Johann Brehmer, Pim de Haan, Phillip Lippe, Taco Cohen: **Weakly supervised causal representation learning**. Thirty-sixth Conference on Neural Information Processing Systems (NeurIPS), 2022 (link).
  — *Own contributions:* Advised in theory and model development, created datasets, implemented causal discovery pipeline, and contributed to writing.
- Phillip Lippe, Sara Magliacane, Sindy Löwe, Yuki M. Asano, Taco Cohen, Efstratios Gavves: **CITRIS: Causal Identifiability from Temporal Intervened Sequences**. International Conference on Machine Learning (ICML), 2022 (link). **[Spotlight]**
- Anna Langedijk, Verna Dankers, Phillip Lippe, Sander Bos, Bryan Cardenas Guevara, Helen Yannakoudakis, and Ekaterina Shutova: **Meta-learning for fast cross-lingual adaptation in dependency parsing**. Proceedings of the 60th Annual Meeting of the Association for Computational Linguistics (ACL, Volume 1: Long Papers), 2022 (link). **[Oral]**
  — *Own contributions:* Advised in training and finetuning of large language Transformers, guided as teaching assistant through the project.
- Phillip Lippe, Taco Cohen and Efstratios Gavves: **Efficient Neural Causal Discovery without Acyclicity Constraints**. International Conference on Learning Representations (ICLR), 2022 (link).
- Phillip Lippe and Efstratios Gavves: **Categorical Normalizing Flows via Continuous Transformations**. International Conference on Learning Representations (ICLR), 2021 (link).

Journals

- Sindy Löwe, Phillip Lippe, Maja Rudolph, Max Welling: **Complex-Valued Autoencoders for Object Discovery**. Transactions on Machine Learning Research, Nov 2022 (link).
  — *Own contributions:* Advised in model development, experiment design and model optimization. Contributed to writing
- Douwe Kiela, Hamed Firooz, Aravind Mohan, Vedanuj Goswami, Amanpreet Singh, Casey A. Fitzpatrick, Peter Bull, Greg Lipstein, Tony Nelli, Ron Zhu, Niklas Muennighoff, Rize Velioglu, Jewgeni Rose, Phillip Lippe, Nithin Holla, Shantanu Chandra, Santhosh Rajamanickam, Georgios Antoniou, Ekaterina Shutova, Helen Yannakoudakis, Vlad Sandulescu, Umut Ozertem, Patrick Pantel, Lucia Specia, Devi Parikh (2021). **The Hateful Memes Challenge: Competition Report.** Proceedings of the NeurIPS 2020 Competition and Demonstration Track, PMLR 133:344-360, 2021 (link).
  — *Own contributions:* Added summary of own solution and results to the Hateful Memes Challenge.

SELECTED PUBLIC TALKS

02/2024 Invited Talk at the Bellairs Workshop on Causality on "On Practical Challenges of Scaling Causal Representation Learning"
09/2023 Invited Talk at the AI4Science Talk Series on "Achieving Accurate Long Rollouts with Neural PDE Solvers"
02/2023 Invited Talk at the Rising Stars in AI Symposium at KAUST on "Causal Representation Learning"
12/2022 Invited Talk at the Google Student Developer Club, University of Augsburg on "Machine Learning with JAX and Flax"
10/2022 Invited Talk at the Innovation Center for Artificial Intelligence on "Causal Representation Learning"
08/2022 Invited Talk at the First Workshop on Causal Representation Learning at UAI 2022 on "Learning Causal Variables from Temporal Sequences with Interventions"
07/2022 Spotlight Talk at ICML 2022 on our paper CITRIS: Causal Identifiability from Temporal Intervened Sequences
07/2022 Contributed Talk at the Workshop On eCommerce at SIGIR 2022
09/2021 Invited Talk at the Innovation Center for Artificial Intelligence on "Neural Causal Discovery"
07/2021 Contributed Talk at the 8th Causal Inference Workshop at UAI 2021
12/2020 Contributed Talk at the NeurIPS 2020 Hateful Memes Competition workshop on our 4th place winning solution

REVIEWING

I have served as reviewer for the following conferences and workshops: ECCV-2020, ICCV-2021, CausalUAI-2021, ICLR-2022, CVPR-2022, CLeaR-2022, NeurIPS-2022, CRL-2022, CML4Impact-2022, CDS-2022, nCSI-2022, ICLR-2023, CLeaR-2023, TSRL4H-2023, Physics4ML-2023, UAI-2023, Frontiers4LCD-2023, NeurIPS-2023.